

MINI PROJECT SYNOPSIS

Reinforcement Learning-Based Autonomous Rocket Landing

Course: V Sem B.Tech. (M&C) Fundamentals of Machine Learning — Lab Batch: Batch 1

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1. Abstract

Domain: Autonomous Control Systems & Applied Reinforcement Learning

Problem Statement: Autonomous vertical landing of reusable rocket boosters represents a high-dimensional, non-linear continuous control challenge where real-time atmospheric dynamics and fuel constraints render conventional hand-tuned control loops suboptimal.

Proposed Solution & MDP Formulation: A custom 2D Gymnasium environment (**RocketLanding-v0**) paired with Deep Q-Networks (DQN) and Deep Deterministic Policy Gradient (DDPG) agents to learn dynamic thrust and orientation control directly from state feedback.

Problem Type: Reinforcement Learning (Markov Decision Process / Sequential Control)

Expected Outcome: A robust RL policy exported to ONNX format, benchmarked on stability and fuel efficiency, and deployed via a serverless HTML5 Canvas + ONNX Runtime Web client-side interactive interface.

2. Dataset Description

- Source & Scale:** Dynamic state-action transition vectors generated via physics simulation in Gymnasium ($\sim 100,000+$ transitions across training episodes).
- Features (6 Continuous Variables):** Horizontal position (x), vertical position (y), linear velocities ($\dot{x}, \dot{y}$), rocket tilt angle (θ), and angular velocity ($\dot{\theta}$).
- Target Variable:** Firing action (a) – control inputs for main thrust throttle and lateral orientation thrusters.
- Assumptions & Limitations:** 2D rigid-body kinematics, simplified drag dynamics, constant gravitational acceleration, and fixed landing pad coordinates.

3. Flowchart of Proposed ML Model

[Gymnasium Kinematics Env] $\rightarrow$ [MDP & Reward Formulation] $\rightarrow$ [DQN / DDPG Training] $\rightarrow$ [Policy Evaluation] $\rightarrow$ [ONNX Export] $\rightarrow$ [HTML5 Canvas Web App]

4. Functional Requirements

- Environment Simulation:** Instantiate 2D rocket physics, engine force vectors, fuel depletion, and boundary detection.
- State Preprocessing & Reward Engine:** Scale state vectors and compute rewards penalizing tilt, velocity, distance, and fuel burn.
- Model Training & Evaluation:** Train Deep Q-Networks with experience replay in PyTorch; log episodic rewards and landing success rates.
- Edge Deployment:** Export final policy weights to `.onnx` and implement client-side inference using ONNX Runtime Web.

5. Machine Learning Models / Algorithms

- Deep Q-Network (DQN):** Maps discrete action sets (thrusters on/off) via deep value function approximation with target networks for stability.
- Deep Deterministic Policy Gradient (DDPG):** Actor-Critic model evaluated for continuous throttle control and precise vectoring.

6. Performance Evaluation

- Landing Success Rate (%):** Percentage of descent episodes concluding within safe impact velocity ($v \leq v_{\max}$) and tilt ($\theta \leq \theta_{\max}$).
- Episodic Metrics:** Average cumulative reward, mean fuel consumption, trajectory landing error ($|x_{\text{final}} - x_{\text{target}}|$), and policy convergence rate.

7. Expected Output

- Interactive, application rendering live autonomous rocket landings driven directly by ONNX web inference.
- Comparative performance logs and convergence plots comparing baseline controllers against DQN/DDPG models.